Lecture 26
Errors, CIs, the sign test

Review

1 What is a simple hypothesis?

one that names a single value of the parameter, such as $H_0: \theta = \theta_0$

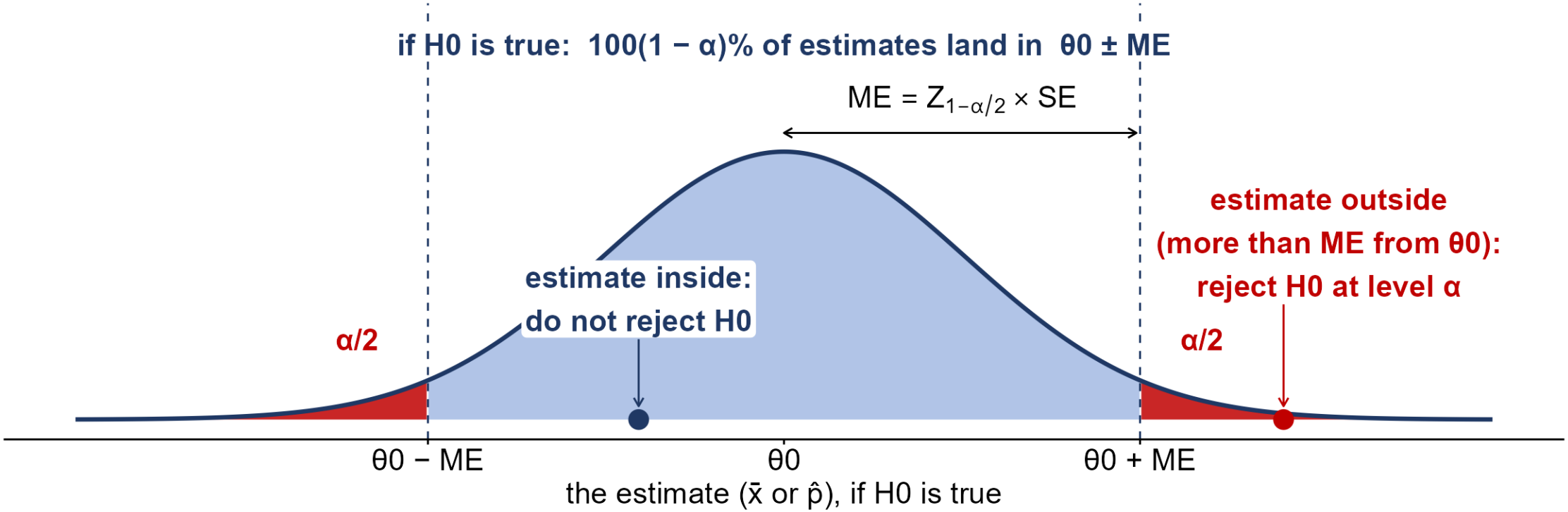
2 What is a composite hypothesis?

one that allows a range of values, such as $H_0: \theta \leq \theta_0$ or $H_0: \theta \geq \theta_0$

3 Where do we test a composite null?

at its boundary θ_0, the value in H_0 that is hardest to reject

Confidence intervals and two-sided tests



the estimate is more than ME = $Z_{1-\alpha/2}$ × SE from θ_0: reject H_0 at level α
the same as: θ_0 lies outside the 100(1 − α)% CI, estimate ± ME

Types of errors

Type I error

reject H_0 when it is true: a false alarm; P(Type I error) = α

Type II error

do not reject H_0 when it is false: a missed effect; P(Type II error) = β

power

reject H_0 when it is false: power = 1 – β

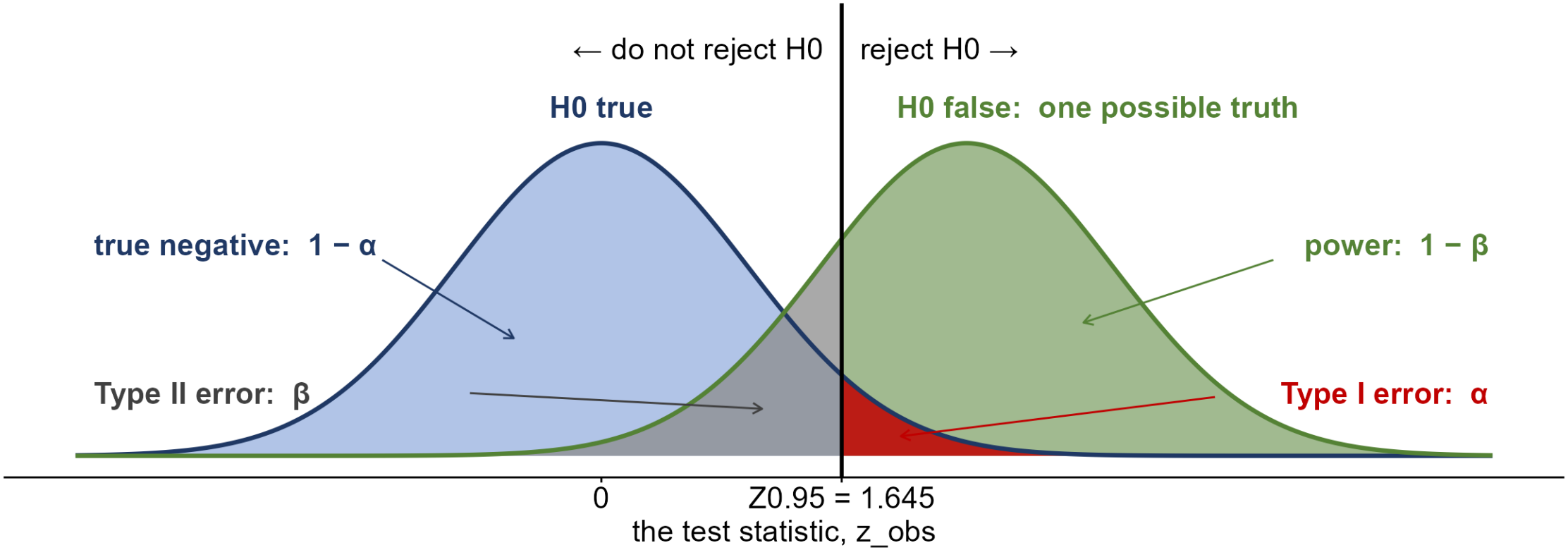
α and β

for a fixed n, making one smaller makes the other larger

Decisions and the truth

	H_0 is true	H_0 is false
Reject H_0	**Type I error** false positive probability α	**Correct decision** true positive probability $1 - \beta$ = power
Do not reject H_0	**Correct decision** true negative probability $1 - \alpha$	**Type II error** false negative probability β

Errors and power on the distributions



Errors in context: climbing equipment

Before a climb, Navah inspects her ropes, harness and anchors. She has used the gear before, so she assumes it is safe, and retires it only if the inspection gives strong evidence that it is not. (OpenStax, Example 9.5)

H_0: Navah's rock climbing equipment is safe

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) Navah thinks her equipment may not be safe when it really is: she retires good gear
(b) Navah thinks her equipment is safe when it is not: she climbs on it
(c) Type II: unsafe gear can fail during the climb

Errors in context: Genetic Labs

A company called Genetic Labs claims its procedure raises the chance that a pregnancy results in a boy. Statisticians test the claim by comparing the share of boys born to its customers with the usual share. (OpenStax, Example 9.7)

H_0: Genetic Labs has no effect on the baby's sex

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) we conclude Genetic Labs changes the odds when it has no effect
(b) we conclude it has no effect when it really does change the odds
(c) Type I: people would pay for a procedure that does nothing

Errors in context: a cure-rate claim

A drug company claims its experimental drug cures at least 75% of men with prostate cancer. Patients and doctors use that figure to choose a treatment, and a research group tests the claim. (OpenStax, Example 9.8)

H_0: $p \geq 0.75$ H_A: $p < 0.75$

(a) Describe a Type I error in context.

(b) Describe a Type II error in context.

(c) Which error has the greater consequence?

(a) we conclude the cure rate is under 75% when it is not: patients may skip a drug that works
(b) we accept the 75% claim when the rate is lower: patients rely on a weaker drug
(c) Type II: the 75% claim drives the choice of treatment

Practice: cell phone owners

Marketers believe 92% of U.S. adults own a cell phone. A manufacturer thinks it is lower and surveys 200 adults. (OpenStax, Try It 9.18)

$n = 200$; 174 own a cell phone; $p_0 = 0.92$; $\alpha = 0.05$

(a) What are H_0 and H_A? Is the sample large enough?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision?
(d) What are the Type I and Type II errors?

(a) $H_0: p \geq 0.92$ $H_A: p < 0.92$; $np_0 = 184$, $n(1 - p_0) = 16$, both ≥ 15
(b) $\hat{p} = 0.87$; $SE = \sqrt{(0.92 \times 0.08 / 200)} \approx 0.0192$; $z_{obs} \approx -2.60$
(c) $P(Z \leq -2.60 \mid p = 0.92) = 0.0047 < 0.05$: reject H_0
(d) I: say fewer than 92% own one when 92%+ do. II: miss that fewer than 92% do

Statistical power

power

P(reject H_0 | H_0 false) = 1 – β

β

P(do not reject H_0 | H_0 false): the chance of missing the effect

power depends on the truth

it is computed at one assumed true value of the parameter

low power

a real effect is easily missed: "do not reject" is not "H_0 is true"

Example: power for the goggles

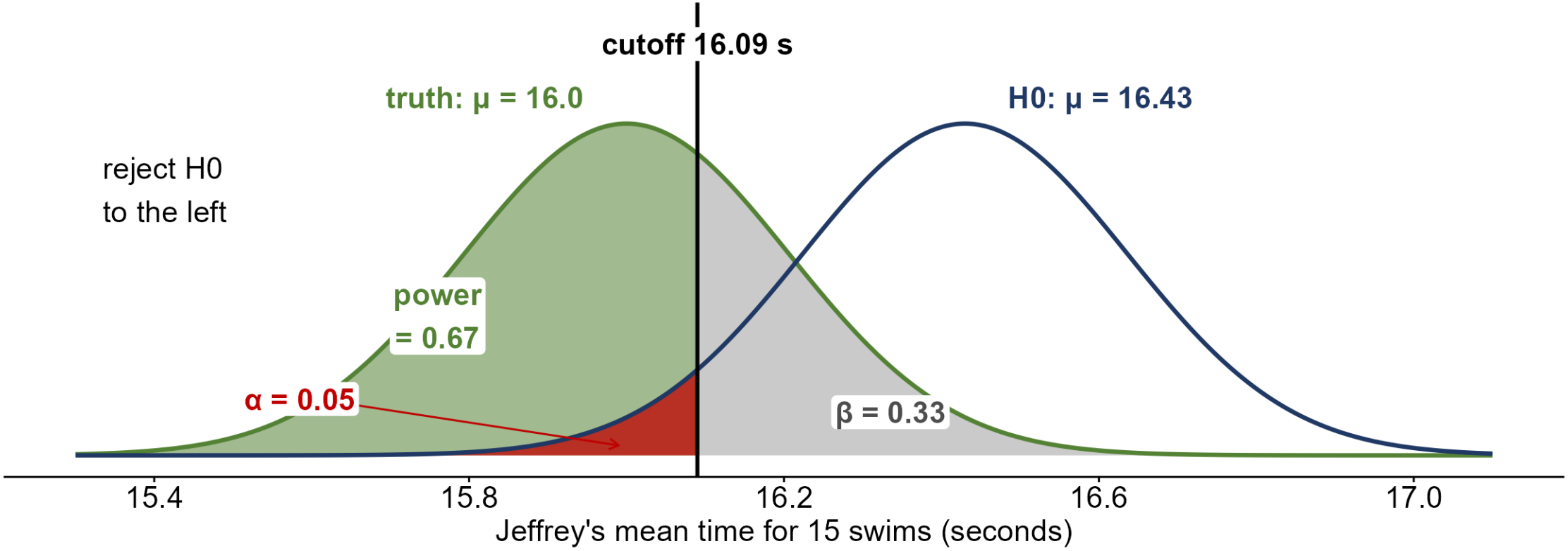
Jeffrey's mean 25-yard freestyle time is 16.43 s, with σ = 0.8. Suppose new goggles really cut his mean to 16.0 s. (OpenStax, Example 9.14)

n = 15; σ = 0.8 s; μ_0 = 16.43; true μ = 16.0; α = 0.05

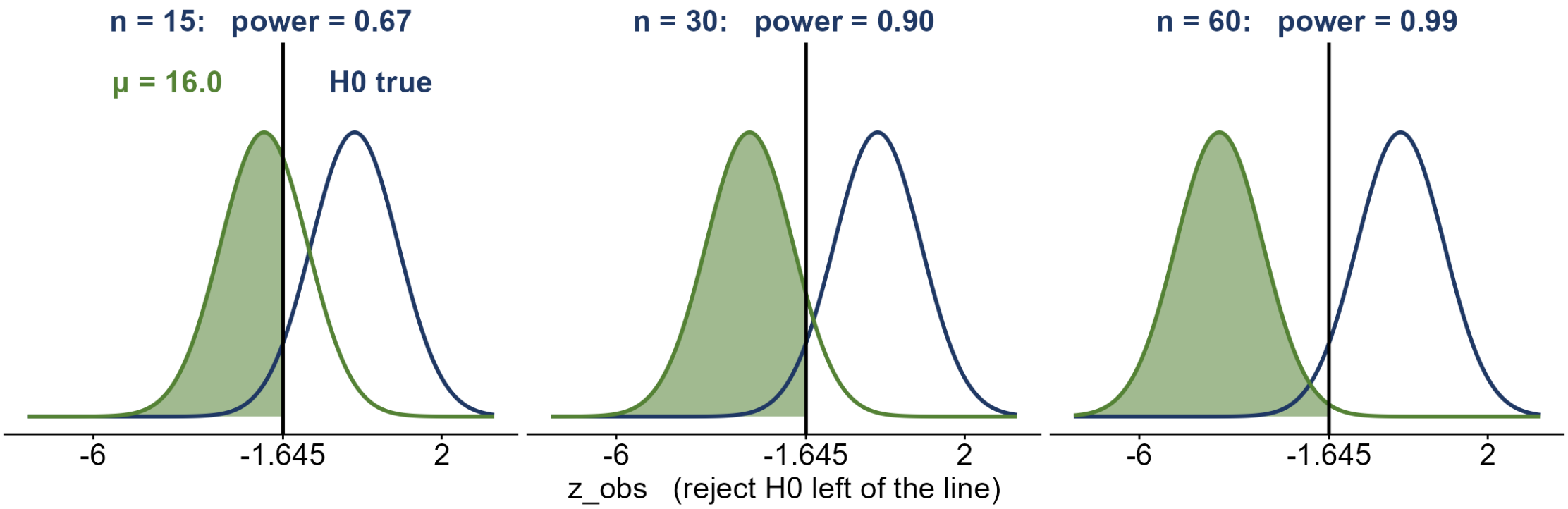
(a) H_A: μ < 16.43. Below what sample mean does the test reject H_0?
(b) If the true mean is 16.0, what is the power?
(c) What is β? What does it mean?

(a) SE = 0.8 / √15 ≈ 0.2066; reject if $\bar{x} \le 16.43 - 1.645(0.2066) \approx 16.09$
(b) $P(\bar{x} \le 16.09 \mid \mu = 16.0) = P(Z \le (16.09 - 16.0) / 0.2066) = P(Z \le 0.44) = 0.67$
(c) β = 1 − 0.67 = 0.33: a third of the time, the test misses the goggles working

α, β and power

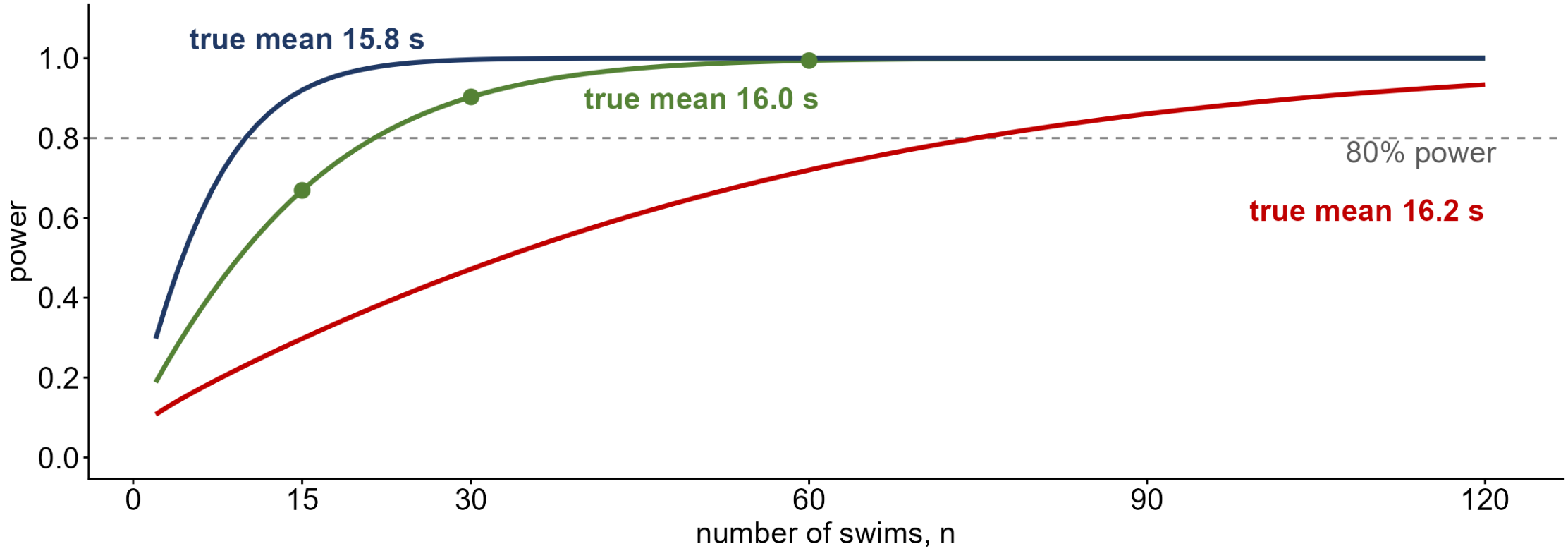

cutoff 16.09 s
truth: μ = 16.0
H0: μ = 16.43
reject H0
to the left
power
= 0.67
α = 0.05
β = 0.33
15.4
15.8
16.2
16.6
17.0
Jeffrey's mean time for 15 swims (seconds)

Power and the test statistic as n grows



the null curve and the cutoff stay put; as n grows, z_{obs} for μ = 16.0 moves away from 0

Power against sample size



with a true mean of 16.0 s, n = 22 swims give 80% power; for 16.2 s it takes n = 75

What raises the power?

more data: n = 30

power 0.67 → 0.90: a smaller SE pulls the two curves apart

a larger $\alpha = 0.10$

power 0.67 → 0.79, but more Type I errors

a bigger effect: $\mu = 15.8$ s

power 0.67 → 0.92: big effects are easier to detect

less variability

a smaller σ shrinks the SE, like a bigger n

Practice: testing water for E. coli

A microbiologist tests a water sample for E. coli before it is approved for drinking. Her test can raise a false alarm or miss bacteria that are there. (OpenStax, Section 9.2, Exercise 19)

H_0: the sample does not contain E. coli

(a) P(she thinks it has E. coli | it does not) = 0.012. Which error, and which letter?
(b) P(she thinks it is clean | it has E. coli) = 0.002. Which error, and which letter?
(c) What is the power of her test? What does it mean?

(a) a Type I error (a false alarm): $\alpha = 0.012$
(b) a Type II error (a miss): $\beta = 0.002$
(c) power = $1 - \beta = 0.998$: she detects E. coli 99.8% of the time it is there

Practice: is the patient sick?

A doctor runs a diagnostic test on a patient. Unless the test gives strong evidence of illness, the doctor treats the patient as healthy and sends them home. (OpenStax, Try It 9.6)

H_0: the patient is not sick

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) the doctor decides a healthy patient is sick: needless worry and treatment
(b) the doctor decides a sick patient is healthy: the patient goes home untreated
(c) Type II: a sick patient goes untreated

Assumptions of the tests so far

every test so far

a random sample (or randomized experiment) of independent observations

z-test for p

Y ~ Bernoulli(p), so the count is Binomial(n, p); $np_0 \geq 15$ and $n(1 - p_0) \geq 15$

z-test for μ (σ known)

the population is approximately normal, $X \sim N(\mu, \sigma)$, or $n \geq 30$ (CLT)

t-test for μ (σ unknown)

the population is approximately normal, $X \sim N(\mu, \sigma)$, or $n \geq 30$ (CLT)

When is a t-test a stretch?

Do husbands and wives see who decides the same way? 17 couples each rate who had more say in a recent major financial decision, from 1 (the wife) to 7 (the husband). Each spouse rates separately, and we look at husband − wife for each couple.

n = 17 couples; ratings 1 to 7; differences from −1 to 5

(a) What would a t-test on the differences assume?
(b) Is that believable here?
(c) What could we use instead?

(a) the differences are a sample from a normal population (or n is large)
(b) doubtful: n = 17 is small, and 1–7 ratings are crude, ordered scores
(c) a test that uses only whether each difference is + or −

Parametric vs non-parametric tests

parametric test

assumes a model for the population, such as $X \sim N(\mu, \sigma)$: the z- and t-tests

non-parametric test

makes few or no assumptions about the population's distribution

advantages

works for small samples, skewed data, ranks and ratings; simple to compute

disadvantage

less power than the parametric test when that test's assumptions hold

The sign test

what it tests

a population median; with matched pairs, the median difference M_D

the signs

write + or − for each difference; drop the zeros; n = the number left

the key fact

half of a population lies above its median: if $M_D = 0$, then $P(+) = 0.5$

the proportion

p = P(a difference is +): H_0: $M_D = 0$ is the same as H_0: p = 0.5

Sign test: the hypotheses

H_0: $M_D = 0$

p = 0.5: no systematic difference; + and − equally likely

H_A: $M_D > 0$

p > 0.5: upper-tailed; the first measurement tends to be larger

H_A: $M_D < 0$

p < 0.5: lower-tailed; the first measurement tends to be smaller

H_A: $M_D \neq 0$

p ≠ 0.5: two-sided; a difference in either direction

Sign test: the test statistic

the test statistic

S = the number of + signs among the n non-zero differences

if H_0 is true

S ~ Binomial(n, 0.5): $P(S = k) = n! / (k! (n - k)!) \times 0.5^n$

mean and SE

$E(S) = n \times 0.5 = n/2$; $SE(S) = \sqrt{(n \times 0.5 \times 0.5)} = \sqrt{n} / 2$

large n ($n/2 \geq 15$)

$z_{obs} = (S - n/2) / (\sqrt{n} / 2) \sim N(0, 1)$: the z-test for p with $p_0 = 0.5$

Sign test: the P-value

H_A: $M_D > 0$ $(p > 0.5)$

P(S ≥ s | H_0 true): add the binomial probabilities from s up to n

H_A: $M_D < 0$ $(p < 0.5)$

P(S ≤ s | H_0 true): add the binomial probabilities from 0 up to s

H_A: $M_D \neq 0$ $(p \neq 0.5)$

2 × the smaller of the two tails

which table?

n/2 < 15: the exact binomial; n/2 ≥ 15: the z-table with z_{obs}

Data: who made the decision?

Couple	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Husband	5	4	6	6	3	2	5	3	1	4	5	4	4	7	5	5	5
Wife	3	3	4	5	3	3	2	3	2	3	2	2	5	2	5	3	1
Difference	2	1	2	1	0	−1	3	0	−1	1	3	2	−1	5	0	2	4

Ratings: 1 = the wife had more say, 7 = the husband had more say. Difference = husband − wife, so + means the husband saw himself as having more say.

Example: who made the decision?

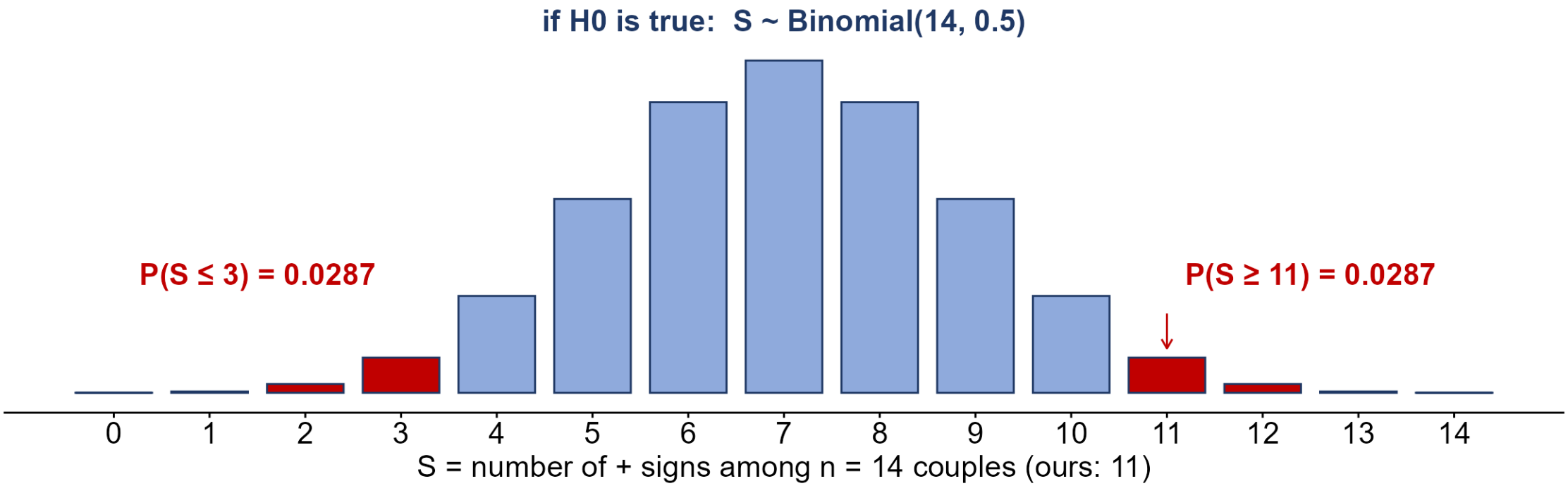
Is there a systematic difference in how husbands and wives see a major financial decision? For each of the 17 couples, the difference is the husband's rating minus the wife's.

2, 1, 2, 1, 0, −1, 3, 0, −1, 1, 3, 2, −1, 5, 0, 2, 4; $\alpha = 0.05$

(a) How many + and − signs are there? What are n and S?
(b) State H_0 and H_A in context.
(c) If H_0 is true, what are E(S) and SE(S)? Can we use z?
(d) Find the P-value. What is the decision?

(a) 11 plus, 3 minus, 3 zeros dropped: n = 14, S = 11
(b) M_D = median of husband − wife: H_0: $M_D = 0$ (p = 0.5), H_A: $M_D \neq 0$
(c) E(S) = 7, SE(S) = $\sqrt{14} / 2 \approx 1.87$; n/2 = 7 < 15: use the exact binomial
(d) $2 \times P(S \geq 11) = 2 \times 0.0287 = 0.0574 > 0.05$: do not reject H_0

The sign test's P-value



$P(S \geq 11) = [C(14, 11) + C(14, 12) + C(14, 13) + C(14, 14)] / 2^{14} = 470 / 16{,}384 = 0.0287$

two-sided: P-value = 2 × 0.0287 = 0.0574

Practice: a stretching program

A physical therapist thinks a six-week stretching program lowers knee pain. Ten patients rate their pain from 0 to 10 before and after the program; the difference is before − after, so + means less pain. (hypothetical data)

3, 1, 0, 2, 4, −1, 2, 1, 3, 2; $\alpha = 0.05$

(a) How many + and − signs are there? What is n?
(b) State H_0 and H_A.
(c) Find the P-value.
(d) What is the decision?

(a) 8 plus, 1 minus, 1 zero dropped: $n = 9$, $S = 8$
(b) $H_0: M_D = 0$ $H_A: M_D > 0$ (upper-tailed; $p > 0.5$)
(c) $P(S \geq 8) = (9 + 1) / 512 \approx 0.0195$
(d) $0.0195 < 0.05$: reject H_0; the program appears to lower pain

Practice: a new coffee blend

A coffee shop thinks its regulars prefer a new blend. 60 regular customers each rate the old and the new blend from 1 to 10; the difference is new − old. 38 are +, 16 are −, and 6 are ties. (hypothetical data)

n = 60 customers; 38 +, 16 −, 6 ties; $\alpha = 0.05$

(a) What are n and S?
(b) State H_0 and H_A.
(c) Find E(S), SE(S) and z_{obs}. Why may we use z here?
(d) Find the P-value. What is the decision?

(a) drop the 6 ties: n = 54, S = 38
(b) H_0: $M_D = 0$ H_A: $M_D > 0$ (upper-tailed; $p > 0.5$)
(c) E(S) = 27, SE(S) = $\sqrt{54} / 2 \approx 3.674$; $n/2 = 27 \geq 15$; $z_{obs} = (38 - 27) / 3.674 \approx 2.99$
(d) $P(Z \geq 2.99) = 1 - 0.9986 = 0.0014 < 0.05$: reject H_0